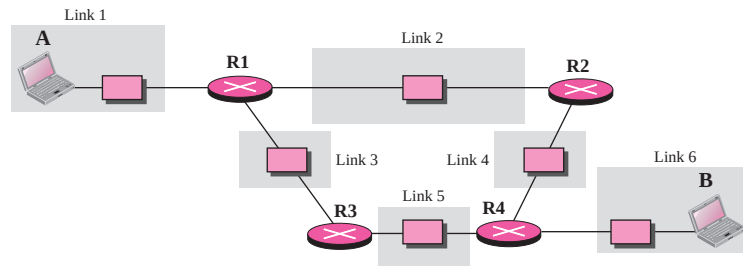


Layers in the TCP/IP Protocol Suite

In this section, we briefly discuss the purpose of each layer in the TCP/IP protocol suite. When we study the purpose of each layer, it is easier to think of a private *internet*, instead of the global Internet. We assume that we want to use the TCP/IP suite in a small, private internet. Such an internet is made up of several small networks, which we call links. A **link** is a network that allows a set of computers to communicate with each other. For example, if all computers in an office are wired together, the connection makes a link. If several computers belonging to a private company are connected via a satellite channel, the connection is a link. A link, as we discussed in Chapter 3, can be a LAN (local area network) serving a small area or a WAN (wide area network) serving a larger area. We also assume that different links are connected together by devices called *routers* or *switches* that route the data to reach their final destinations. Figure 2.9 shows our imaginary internet that is used to show the purpose of each layer. We have six links and four routers (R1 to R4). We have shown only two computers, A and B.

Figure 2.9 A private internet



Physical Layer

TCP/IP does not define any specific protocol for the physical layer. It supports all of the standard and proprietary protocols. At this level, the communication is between two hops or nodes, either a computer or router. The unit of communication is a single bit. When the connection is established between the two nodes, a stream of bits is flowing between them. The physical layer, however, treats each bit individually. Figure 2.10 shows the communication between nodes. We are assuming that at this moment the two computers have discovered that the most efficient way to communicate with each other is via routers R1, R3, and R4. How this decision is made is the subject of some future chapters.

Note that if a node is connected to n links, it needs n physical-layer protocols, one for each link. The reason is that different links may use different physical-layer protocols. The figure, however, shows only physical layers involved in the communication. Each computer involves with only one link; each router involves with only two links. As Figure 2.10 shows, the journey of bits between computer A and computer B is made of four independent short trips. Computer A sends each bit to router R1 in the format of the protocol used by link 1. Router 1 sends each bit to router R3 in the format dictated by the protocol used by link 3. And so on. Router R1 has two three physical layers (two are shown in our scenario). The layer connected to link 1 receives bits according to the format of the protocol